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Computer Network (CO3093)

Assignment 2

“NETWORK DESIGN AND SIMULATION FOR A CRITICAL LARGE HOSPITAL”

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HO CHI MINH CITY, NOVEMBER 2025



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Member list & Workload

No.	Fullname	Student ID	Work	% done
1	Nguyễn Duy Thành	2353101	- Configured DHCP, switch settings, IP addressing and check the system.	100%
2	Đặng Sinh Hùng	2352420	- Implemented IT servers and performed Section 3 calculations.	100%
3	Châu Kiến Toàn	2353192	- Research related applications/systems	100%
4	Phạm Quang Tiến Thành	2353103	- Built Main Site (A+B), configured all WLANs, and ran Section 5 tests.	100%

Table 1: Member list & workload

1 Suitable Network Structure

This section analyzes the overall requirements of the hospital's network system, as specified in the assignment brief. It covers the needs for the Main Site, the two Auxiliary Sites, expected data workloads, and future growth projections. Based on this analysis, we will then survey the installation locations, identify high-load areas, and select an appropriate network architecture.

1.1 Network System Requirements

Based on the case study provided, the key IT and network requirements for the Specialized Hospital are broken down by location and function.

1.1.1 Main Site (Ho Chi Minh City)

The Main Site is the central hub of the hospital's operations and has the most extensive requirements:

- **Physical Infrastructure:** Consists of two buildings (A and B), each with 5 floors and 10 rooms per floor.
- **Data Center:** A separate Data Center, IT, and Cabling Central Local room is located 50 meters from buildings A and B.
- **Scale:** Classified as medium-scale, required to support 600 workstations, 10 servers, and at least 12 networking devices.
- **Wireless Coverage:** A comprehensive wireless connection must be available throughout the entire site.
- **Technology Standards:** The network must utilize new technologies, including fiber cabling (GPON) and high-speed Giga Ethernet (1GbE/10GbE/40GbE) for both wired and wireless connections.
- **Network Segmentation:** The network must be organized using a VLAN structure to separate traffic for different hospital departments.
- **WAN Connection:** Must connect to the two auxiliary sites (DBP and BHTQ) via two leased lines, with potential application of SD-WAN or MPLS.
- **Internet Access:** Requires 2 xDSL lines for Internet access, configured with a load-balancing mechanism. All Internet traffic from all sites must pass through this main site's subnet.
- **Core Requirements:** The design must prioritize high availability (HA), robustness, security (e.g., firewall, IPS/IDS), ease of extension, and simple upgrading.
- **Security Services:** The design must propose configurations for both site-to-site VPN and a teleworker VPN, as well as a surveillance camera system.

1.1.2 Auxiliary Sites (DBP Street & BHTQ Street)

The two auxiliary sites are smaller branches that connect back to the Main Site:

- **Physical Infrastructure:** Each site has one building with 2 floors. The first floor contains an IT room and a Cabling Central Local.
- **Scale:** Classified as small-scale, required to support 60 workstations, 2 servers, and 5 or more networking devices per site.
- **WAN Connection:** Must implement a stable WAN link back to the Main Site using technologies like SD-WAN or MPLS.

1.1.3 Data Flow and Workload

The data flow and workload estimates are shared across all three sites:

- **Peak Hours:** System load is expected to be at 80% during peak hours (9:00-11:00 and 15:00-16:00).
- **Server Traffic:** Servers (for updates, web, database) are estimated to have a total traffic of ≈ 1000 MB/day download and ≈ 2000 MB/day upload.
- **Workstation Traffic:** Each workstation (for web browsing, documents, transactions) is estimated at ≈ 500 MB/day download and ≈ 100 MB/day upload.
- **Guest WiFi Traffic:** WiFi-connected devices from customers are estimated to download ≈ 500 MB/day.

1.1.4 Future Growth

The design must be scalable to accommodate future expansion:

- **Growth Rate:** The hospital network is projected to grow by 20% over the next 5 years, impacting the number of users, network load, and potential site extensions.
- **Design Implication:** This growth factor necessitates a **scalable and modular** design. The chosen architecture must support adding new users, devices, and services easily, ensuring that bandwidth and hardware (like switches and routers) can be upgraded without a complete system redesign.

1.2 Checklist to be Surveyed at Installation Locations

Before designing the network, a physical survey of the installation sites is crucial. The following checklist outlines the key information gathered from a hypothetical site survey, based on the project requirements.

1.2.1 Main Site (Buildings A, B, and Data Center)

- **Building Layout:** Two buildings (A & B), 5 floors each. 10 rooms/floor. Total of ≈ 100 rooms.
- **Workstation Density:** ≈ 600 workstations total, averaging 60 workstations per floor (or 6 per room).
- **Cabling Infrastructure:**
 - Horizontal Cabling: Assume existing structured cabling (e.g., Cat 6/6A) from wall plates in each room to a telecommunications room (TR) or wiring closet on each floor.
 - Vertical Cabling (Backbone): Fiber optic cabling is required to connect each floor's TR back to the main distribution frame (MDF).
 - Data Center Link: A high-speed fiber backbone (e.g., 10GbE/40GbE) is required to link the MDFs in Buildings A and B to the main Data Center, IT, and Cabling Central room, located 50m away.
- **Wireless Access:** Full coverage required. This implies at least 2-3 Lightweight Access Points (LWAPs) per floor, per building, to handle density and coverage, managed by a central Wireless LAN Controller (WLC).
- **Redundancy:** Due to High Availability (HA) requirements, redundant Layer 3 switches will be planned for the distribution layer of each building, and redundant core routers/firewalls in the Data Center.

1.2.2 Auxiliary Sites (DBP & BHTQ)

- **Building Layout:** One building per site, 2 floors each.
- **Workstation Density:** ≈ 60 workstations per site.
- **Cabling Infrastructure:**
 - Ground Floor: Contains the IT room and Cabling Central. This will serve as the Main Distribution Frame (MDF) for the site.
 - Second Floor: Requires horizontal cabling to workstations, connecting back to the MDF on the first floor.
- **Redundancy:** While smaller, HA is still important. Redundant links from the access switches to the distribution switches are recommended.

1.3 Areas with High Load

To ensure network performance and stability, it is critical to identify potential bottlenecks. Load balancers and high-capacity equipment will be designated for these areas:

- **Internet Gateway (Main Site):** The two 2xDSL lines must handle all Internet traffic for all 3 sites (600+ workstations + 60 + 60). A load-balancing mechanism is explicitly required here to distribute traffic and provide failover.
- **Data Center / Server Farm:** The 10 servers at the Main Site (handling HIS, RIS-PACS, databases) will serve 720+ workstations. The connections to this server farm (especially the aggregation switches) will experience constant, heavy traffic.
- **WAN Edge (Main Site):** The router/SD-WAN appliance managing the 2 leased lines to the auxiliary sites is a critical choke point. All inter-site data, including software updates and database access, must pass through here.
- **DMZ Zone:** The assignment requires a secure partition for servers (like a Web Server or Mail Server) that are accessible from the Internet. This zone will receive high volumes of untrusted traffic and must be protected by a firewall.

1.4 Chosen Network Structure

To meet the requirements of scalability, security, and high availability, we will design the hospital network based on the industry-standard **Hierarchical Network Topology** (often called the 3-Layer Model). This model segments the network into distinct layers, each with a specific function.

- **Access Layer:** This is the first layer where end devices (workstations, printers, medical devices, Access Points) connect to the network. In our design, this will be implemented using Layer 2 switches (e.g., Cisco Catalyst 2960) located in the wiring closet on each floor of every building.
- **Distribution Layer:** This layer aggregates the traffic from all Access Layer switches on a floor or in a building. It acts as the boundary for broadcast domains and is where routing between VLANs (Inter-VLAN routing) and policy enforcement (Access Control Lists - ACLs) are implemented. This layer will use redundant Layer 3 switches (e.g., Cisco Catalyst 3650) to ensure high availability.
- **Core Layer:** This layer is the high-speed backbone of the network. Its sole purpose is to forward traffic between Distribution Layer devices as fast as possible. This layer connects the Main Site's buildings (A, B, and Data Center) and the WAN edge. High-speed (10G/40G) Layer 3 switches or routers are used here, also in a redundant configuration.

This hierarchical design provides significant benefits:

- **Scalability:** As the hospital grows 20%, new access switches or even new buildings (new distribution blocks) can be added without redesigning the entire network.
- **Redundancy & Availability (HA):** By deploying redundant switches at the Core and Distribution layers, the network can withstand a device failure without a service interruption.
- **Performance:** Traffic is segmented. Local traffic stays local (within the Access layer), while Inter-VLAN traffic is efficiently routed at the Distribution layer, preventing the Core from becoming a bottleneck.
- **Security & Manageability:** The layered structure makes it easy to implement security policies (ACLs, Firewalls) at the Distribution layer, controlling traffic flow between different departments (VLANs).

1.5 Wireless, Security, and Server Partitions

To fulfill the requirements for security and mobility, the network design must incorporate specific zones for servers and a comprehensive wireless strategy.

1.5.1 Wireless Network Design

A centrally-managed wireless network will be implemented to meet the requirement for full wireless coverage.

- **Technology:** A controller-based architecture will be used, deploying **Lightweight Access Points (LWAPs)** on each floor. These LWAPs will be managed by a central **Wireless LAN Controller (WLC)** located in the Main Site's Data Center. This simplifies configuration, monitoring, and roaming.
- **Security:** The wireless network will be secured using **WPA3 (or WPA2-Enterprise)** standards, requiring users to authenticate via a central RADIUS server (AAA) to ensure only authorized personnel can access the internal network.
- **Guest Access:** A separate "Guest" SSID will be created for customers. This network will be isolated on its own VLAN and will be blocked from accessing any internal hospital resources (as required in Step 5), with traffic routed directly to the Internet.

1.5.2 Network Partitions (Server Farm, DMZ, Firewall)

To secure data and meet high-security requirements, the network will be segmented into distinct security zones using a firewall:

- **Server Farm (Internal):** This zone will be located in the Data Center and is part of the trusted internal network. It hosts the 10 critical internal servers (HIS, RIS-PACS, LIS, databases). Access to this zone will be strictly controlled by the firewall and ACLs.
- **DMZ (Demilitarized Zone):** As seen in the reference diagrams, a DMZ partition is required. This is a semi-trusted zone, isolated from the internal network by the firewall. It will host public-facing servers, such as the hospital's Web Server, Mail Server, or DNS Server, which need to be accessible from the Internet.
- **Firewall:** A **Next-Generation Firewall (NGFW)** will be the primary security appliance. It will be placed at the edge of the network (at the Main Site), inspecting all traffic flowing between the Internet, the DMZ, and the Internal Network (Server Farm, Buildings A & B, and the WAN links to auxiliary sites). This placement is critical as all traffic must pass through the main site.

2 List of minimum equipment, IP plan, and wiring diagram (cabling)

This section specifies the minimum equipment set, an IP addressing plan that enables summarization and scalable inter-VLAN routing, and the corresponding wiring principles. All selected devices are available in Cisco Packet Tracer to ensure the design can be demonstrated end-to-end.

2.1 Recommended equipment and typical specifications

The following items present the Packet Tracer-compatible equipment used in the design. Each subsection explains the device's role in full sentences, lists key characteristics using complete statements, and includes a placeholder image that you can replace later. The order follows the logical flow of traffic from the core through the access layer and perimeter to the WAN edge and wireless services, and then it mirrors the same pattern at the auxiliary sites to maintain operational consistency.

2.2 Switches

2.2.1 Cisco Catalyst 3650-24PS (Core Layer 3 Switch)

At the Main Site, the Catalyst 3650-24PS acts as the Layer 3 core and default gateway for most VLANs, providing inter-VLAN routing and policy enforcement.

- **Role.** Terminates campus VLANs (user, server, management, guest) and routes between them; the DMZ gateway can be moved to the firewall if stricter isolation is required.
- **Interfaces.** Twenty-four FastEthernet access ports plus GigabitEthernet uplinks used as 802.1Q trunks toward access switches and upstream devices.
- **Key features.** SVIs for inter-VLAN routing, OSPF for internal routing, ACLs for segmentation, DHCP relay, VLANs, trunking, and Spanning Tree.



Figure 1: Catalyst 3650-24PS at the core (placeholder).

2.2.2 Cisco Catalyst 2960-24TT (Access Layer Switch)

The Catalyst 2960-24TT provides wired edge connectivity and extends the VLAN design down to end hosts.

- **Role.** Aggregates PCs and other wired clients on each floor and forwards their traffic to the core over a tagged trunk.
- **Interfaces.** Twenty-four Ethernet access ports for clients plus uplink interfaces used as 802.1Q trunks to the core.

- **Key features.** VLAN access ports, 802.1Q trunking, Spanning Tree for loop prevention, and Port Security for basic edge hardening.



Figure 2: Catalyst 2960-24TT at the access layer (placeholder).

2.3 Security Devices

2.3.1 Cisco ASA (Firewall)

The Cisco ASA firewall forms the security perimeter between Inside, DMZ, and Outside zones.

- **Role.** Forwards Internet-bound traffic toward the ISP, protects DMZ services, and enforces separation between internal and external networks.
- **Interfaces and zones.** Inside, DMZ, and Outside interfaces with distinct security levels and policies.
- **Key features.** NAT/PAT, ACL-based firewalling, static and default routing, and optional site-to-site IPsec tunnels for secure inter-site links.



Figure 3: ASA at the perimeter with Inside/DMZ/Outside interfaces (placeholder).

2.4 Routers and WAN Edge

2.4.1 Cisco 2811 (WAN Edge Router of Auxiliary Sites)

The Cisco 2811 operates as the WAN edge, terminating point-to-point links to the auxiliary sites.

- **Role.** Establishes serial /30 links to DBP and BHTQ, exchanges routes with the Main Site, and forwards traffic toward the ASA for centralized Internet access.
- **Interfaces.** FastEthernet for LAN/ASA connectivity and serial interfaces for leased-line style connections to remote sites.
- **Key features.** OSPF or static routing, default routes, ACLs for simple filtering, and basic NAT when a dedicated firewall is not used at a branch.



Figure 4: Cisco 2811 as WAN edge (placeholder).

2.4.2 Cloud-PT and DSL-Modem-PT (ISP Emulation)

The ISP environment is emulated with Cloud-PT and DSL-Modem-PT objects.

- **Role.** Provide simulated upstream connectivity so that default routes, primary/backup Internet paths, and failover behavior can be demonstrated.
- **Interfaces.** Cloud-PT exposes Ethernet, Serial, and DSL interfaces; each DSL modem offers a DSL-facing port and an Ethernet handoff toward the ASA or WAN router.
- **Dual-ISP.** Dual modems are used to model dual-ISP designs with active-standby default routes or adjusted administrative distances.

2.5 Wireless Infrastructure

2.5.1 Access Point Cisco 3702I-C-K9 (LWAP)

Cisco 3702I-C-K9 access points extend the campus VLAN and policy model to wireless clients.

- **Role.** Broadcast staff and guest SSIDs and map them to the corresponding VLANs so that wireless users follow the same segmentation as wired clients.
- **Uplink.** A single Ethernet uplink toward the access switch carries tagged traffic for multiple SSIDs/VLANs.
- **Key features.** Multiple SSIDs, basic security (e.g., pre-shared keys) for the lab, and VLAN tagging on the uplink.



Figure 5: Autonomous AccessPoint-PT serving staff and guest SSIDs (placeholder).

2.6 Infrastructure Servers

2.6.1 Server-PT (Infrastructure Services)

Server-PT nodes host core services in the server VLAN at the Main Site.

- **Role.** Provide DHCP, DNS, and NTP for clients and devices, and optionally emulate web or AAA services for DMZ testing.
- **Connectivity.** Single Ethernet interface in the server VLAN, using the core SVI as default gateway.
- **Key features.** Simple graphical configuration of address pools, DNS zones, and time settings to support lab validation.

2.7 Auxiliary / Branch Sites (DBP and BHTQ)

2.7.1 Auxiliary Sites (DBP and BHTQ)

Each auxiliary site reuses a scaled-down version of the Main Site architecture.

- **Branch core.** A Catalyst 3650-24PS terminates local VLANs and exchanges routes with the Main Site.
- **Access layer.** Catalyst 2960-24TT switches connect end devices and uplink to the branch core via 802.1Q trunks.
- **WAN edge.** A Cisco 2811 provides the point-to-point WAN link back to the Main Site using /30 addressing.
- **Wireless.** Autonomous access points extend staff and guest SSIDs to wireless clients at the branch.
- **Security.** Basic ACL/NAT can be done on the 2811, or an ASA can be added where stronger perimeter separation is required.



2.8 IP plan

2.8.1 Main site

Table 2: Building A IP Addressing

Building A IP PLAN						
VLAN	Name	Network & Subnet	Valid IP Range	Default GW	Broadcast	WLC
10	PC-Floor1A	192.168.1.0/24	192.168.1.2–192.168.1.253	192.168.1.1	192.168.1.254	None
20	PC-Floor2A	192.168.2.0/24	192.168.2.2–192.168.2.253	192.168.2.1	192.168.2.254	None
30	PC-Floor3A	192.168.3.0/24	192.168.3.2–192.168.3.253	192.168.3.1	192.168.3.254	None
40	PC-Floor4A	192.168.4.0/24	192.168.4.2–192.168.4.253	192.168.4.1	192.168.4.254	None
50	PC-Floor5A	192.168.5.0/24	192.168.5.2–192.168.5.253	192.168.5.1	192.168.5.254	None
100	WLC-A	192.168.100.0/24	192.168.100.2–192.168.100.253	192.168.100.1	192.168.100.254	192.168.100.15
200	SecurityCamera	192.168.200.0/24	192.168.200.2–192.168.200.253	192.168.200.1	192.168.200.254	None

Table 3: Building B IP Addressing

Building B IP PLAN						
VLAN	Name	Network & Subnet	Valid IP Range	Default GW	Broadcast	WLC
10	PC-Floor1B	192.168.6.0/24	192.168.6.2–192.168.6.253	192.168.6.1	192.168.6.254	None
20	PC-Floor2B	192.168.7.0/24	192.168.7.2–192.168.7.253	192.168.7.1	192.168.7.254	None
30	PC-Floor3B	192.168.8.0/24	192.168.8.2–192.168.8.253	192.168.8.1	192.168.8.254	None
40	PC-Floor4B	192.168.9.0/24	192.168.9.2–192.168.9.253	192.168.9.1	192.168.9.254	None
50	PC-Floor5B	192.168.10.0/24	192.168.10.2–192.168.10.253	192.168.10.1	192.168.10.254	None
100	WLC-B	192.168.101.0/24	192.168.101.2–192.168.101.253	192.168.101.1	192.168.101.254	192.168.101.15
200	SecurityCamera	192.168.201.0/24	192.168.201.2–192.168.201.253	192.168.201.1	192.168.201.254	None

Table 4: IT Block IP Addressing

IT IP PLAN						
VLAN	Name	Network & Subnet	Valid IP Range	Default GW	Broadcast	WLC
10	Server	192.168.12.0/24	192.168.12.2–192.168.12.253	192.168.12.1	192.168.12.254	None
20	IT	192.168.11.0/24	192.168.11.2–192.168.11.253	192.168.11.1	192.168.11.254	None
100	WLAN	192.168.102.0/24	192.168.102.2–192.168.102.253	192.168.102.1	192.168.102.254	192.168.102.15
200	SecurityCamera	192.168.202.0/24	192.168.202.2–192.168.202.253	192.168.202.1	192.168.202.254	None

2.9 Auxiliary Sites

Table 5: Dien Bien Phu Site IP Addressing

Dien Bien Phu Site IP PLAN						
VLAN	Name	Network & Subnet	Valid IP Range	Default GW	Broadcast	WLC
10	Server	192.168.15.0/24	192.168.15.2–192.168.15.253	192.168.15.1	192.168.15.254	None
20	Room	192.168.14.0/24	192.168.14.2–192.168.14.253	192.168.14.1	192.168.14.254	None
100	WLAN	192.168.102.0/24	192.168.102.2–192.168.102.253	192.168.102.1	192.168.102.254	192.168.102.15
200	SecurityCamera	192.168.203.0/24	192.168.203.2–192.168.203.253	192.168.203.1	192.168.203.254	None

Table 6: Ba Huyen Thanh Quan Site IP Addressing

Ba Huyen Thanh Quan Site IP PLAN						
VLAN	Name	Network & Subnet	Valid IP Range	Default GW	Broadcast	WLC
10	Server	192.168.17.0/24	192.168.17.2–192.168.17.253	192.168.17.1	192.168.17.254	None
20	Room	192.168.16.0/24	192.168.16.2–192.168.16.253	192.168.16.1	192.168.16.254	None
100	WLAN	192.168.103.0/24	192.168.103.2–192.168.103.253	192.168.103.1	192.168.103.254	192.168.103.15
200	SecurityCamera	192.168.204.0/24	192.168.204.2–192.168.204.253	192.168.204.1	192.168.204.254	None

2.13 SD-WAN

Inter-site connectivity is summarized in the SD-WAN diagram. RouterA and RouterB represent the Main Site buildings (A and B), while RouterDBP and RouterBHTQ represent the two Auxiliary Sites. Each edge router forms a WAN underlay link (shown as leased-line connections) to the central *SD-WAN Cloud*. On top of these point-to-point links, SD-WAN tunnels are established so that routes between sites can be managed centrally by policy rather than by individual static configuration. The Main Site advertises a default route toward the ASA for centralized Internet breakout, and each Auxiliary Site advertises its local VLAN aggregates. The internal LAN wiring at each site is unchanged; only the WAN portion is abstracted by the SD-WAN overlay to provide path selection, resilience and easier management.

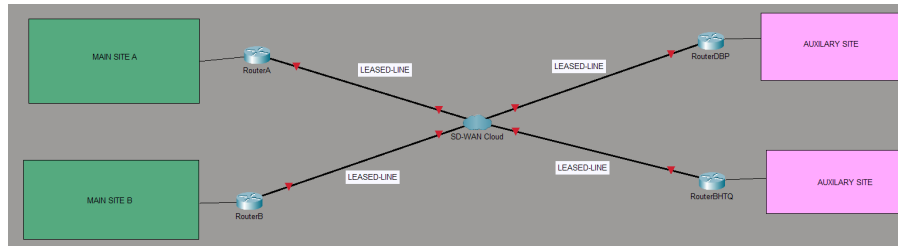


Figure 8: SD-WAN overview connecting Main and Auxiliary Sites

3 Calculate Throughput and Bandwidth and Hospital Network Configuration Suggestion

The hospital network experiences peak traffic at 80% during two daily periods (09:00–11:00 and 15:00–16:00), with total peak duration of 3 hours. Network usage characteristics are as follows:

- Total server traffic: 3000 MB/day (1000 MB download + 2000 MB upload)
- Per workstation traffic: 600 MB/day (500 MB download + 100 MB upload)
- Customer WiFi traffic: 500 MB/day (download)
- Main site: 600 workstations
- Two auxiliary sites: 120 workstations total (60 each)

Total number of workstations in the hospital network: $600 + 120 = 720$

3.1 Main Site (600 Workstations)

Throughput Calculations

- $Throughput_{servers} = \frac{3000 \times 10}{24 \times 3600} = \frac{30000}{86400} = \frac{25}{72} \approx 0.35$ (MB/s).
- $Throughput_{workstations} = \frac{600 \times 600}{24 \times 3600} = \frac{360000}{86400} = \frac{25}{6} \approx 4.17$ (MB/s).
- $Throughput_{WiFi} = \frac{500}{24 \times 3600} = \frac{500}{86400} = \frac{5}{864} \approx 0.006$ (MB/s).

Peak Bandwidth (80% in 3 Hours)

- $Bandwidth_{servers} = \frac{3000 \times 0.8}{3 \times 3600} = \frac{2400}{10800} = \frac{2}{9} \approx 0.22$ (MB/s).
- $Bandwidth_{workstations} = \frac{600 \times 600 \times 0.8}{3 \times 3600} = \frac{288000}{10800} = \frac{80}{3} \approx 26.67$ (MB/s).
- $Bandwidth_{WiFi} = \frac{500 \times 0.8}{3 \times 3600} = \frac{400}{10800} = \frac{1}{27} \approx 0.037$ (MB/s).
- **Total Throughput_{main}** = $0.35 + 4.17 + 0.006 = 4.53$ (MB/s).
- **Total Bandwidth_{main}** = $0.22 + 26.67 + 0.037 = 26.93$ (MB/s).

3.2 Two Auxiliary Sites (120 Workstations)

Throughput Calculations

- $Throughput_{servers} = \frac{3000 \times 2}{24 \times 3600} = \frac{6000}{86400} = \frac{5}{72} \approx 0.069$ (MB/s).
- $Throughput_{workstations} = \frac{120 \times 600}{24 \times 3600} = \frac{72000}{86400} = \frac{5}{6} \approx 0.83$ (MB/s).
- $Throughput_{WiFi} = \frac{500}{24 \times 3600} = \frac{5}{864} \approx 0.006$ (MB/s).

Peak Bandwidth (80% in 3 Hours)

- $Bandwidth_{servers} = \frac{2 \times 3000 \times 0.8}{3 \times 3600} = \frac{4800}{10800} = \frac{4}{9} \approx 0.44$ (MB/s).
- $Bandwidth_{workstations} = \frac{120 \times 600 \times 0.8}{3 \times 3600} = \frac{57600}{10800} = \frac{16}{3} \approx 5.33$ (MB/s).
- $Bandwidth_{WiFi} = \frac{500 \times 0.8}{3 \times 3600} = \frac{1}{27} \approx 0.037$ (MB/s).
- **Total Throughput_{aux}** = $0.069 + 0.83 + 0.006 = 0.905$ (MB/s).
- **Total Bandwidth_{aux}** = $0.44 + 5.33 + 0.037 = 5.81$ (MB/s).

3.3 Total Hospital Bandwidth Requirement

$$TotalBandwidth = 26.93 + 5.81 = 32.74 \text{ (MB/s)}.$$

Convert to Mbps:

$$32.74 \times 8 = 261.92 \text{ Mbps}$$

Accounting for 20% future growth and redundancy, the recommended ISP bandwidth capacity is:

400–500 Mbps

3.4 Hospital Network Configuration Suggestion

Based on the calculated peak requirement of approximately 262 Mbps and the recommended provisioning capacity of 400–500 Mbps, the following configuration aligns with the hierarchical design, IP scheme, and device roles presented.

Internet and WAN Connectivity

- Primary Internet link at the Main Site: 500 Mbps fiber.
- Secondary WAN link: dual xDSL lines for load-sharing and failover, as modeled in Packet Tracer.
- Inter-site connectivity: point-to-point links using Cisco 2811 routers with **/30 addressing and OSPF**, matching the academic model.
- SD-WAN concepts may be referenced for scalability, but the simulated topology uses **serial P2P links with OSPF and default-route centralization**.

Core and Distribution Switching

- Main Site core: **Catalyst 3650-24PS** acting as multilayer switch, terminating SVIs and running OSPF.
- Access layer: **Catalyst 2960-24TT** switches per floor, connected with 802.1Q trunks.
- Uplinks: GigabitEthernet in Packet Tracer simulation; fiber in real deployment as specified in Section 2.3.
- High availability: dual uplinks toward the MDF and redundant paths where possible.

Wireless Infrastructure

- Autonomous **Cisco AccessPoint-PT** nodes per floor, consistent with Section 2.1.6.
- Separate SSIDs for staff and guests mapped to VLANs; WPA2/WPA3 in real deployment, PSKs used for lab simplicity.
- External power in simulation; PoE planned for physical environment.

Addressing and VLAN Design

Following the Section 2.2 IP scheme, all subnets are taken from the private **192.168.0.0/16** block and are consistently split per building and site:

- **Main Site (Buildings A, B and IT Block)**
 - **Building A user VLANs:** 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24, 192.168.4.0/24, 192.168.5.0/24.

- **Building B user VLANs:** 192.168.6.0/24, 192.168.7.0/24, 192.168.8.0/24, 192.168.9.0/24, 192.168.10.0/24.
- **IT Block / Server and IT staff:** 192.168.12.0/24 (Server VLAN) and 192.168.11.0/24 (IT VLAN).
- **WLAN VLANs at the main campus:** 192.168.100.0/24 (Building A), 192.168.101.0/24 (Building B), 192.168.102.0/24 (IT Block).
- **SecurityCamera/IoT VLANs:** 192.168.200.0/24 (Building A), 192.168.201.0/24 (Building B), 192.168.202.0/24 (IT Block).
- **Auxiliary Site – Dien Bien Phu (DBP)**
 - Server VLAN: 192.168.15.0/24.
 - Room/User VLAN: 192.168.14.0/24.
 - WLAN VLAN: 192.168.102.0/24.
 - SecurityCamera VLAN: 192.168.203.0/24.
- **Auxiliary Site – Ba Huyen Thanh Quan (BHTQ)**
 - Server VLAN: 192.168.17.0/24.
 - Room/User VLAN: 192.168.16.0/24.
 - WLAN VLAN: 192.168.103.0/24.
 - SecurityCamera VLAN: 192.168.204.0/24.

Security Architecture

- Edge security: **ASA firewall** at Main Site, as defined in the existing topology.
- DMZ segmentation for public-facing services.
- OSPF route summarization and default-route injection toward the ASA.
- IPSec optional for secure simulation of encrypted WAN tunnels.

Server Infrastructure

- Core services: **DHCP, DNS, NTP** on Server-PT nodes in the Server VLAN.
- AAA and web/file services added as needed for demonstration scenarios.
- PACS/HIS/RIS workloads assumed to run in data center with scalable storage.

Surveillance System

- Dedicated CCTV / SecurityCamera VLANs are mapped as 192.168.200.0/24 (Building A), 192.168.201.0/24 (Building B), 192.168.202.0/24 (IT Block), 192.168.203.0/24 (DBP), and 192.168.204.0/24 (BHTQ).
- IP cameras are powered through PoE in the real design; external power is used in Packet Tracer.
- Video storage is located in the Main Site server room.

High Availability and Reliability

- Dual-provider Internet architecture.
- UPS and generator support for Main Site racks.
- OSPF with summarization to maintain scalable routing tables.
- VLAN segmentation and ACL enforcement to contain broadcast and threat domains.



Management and Monitoring

- SNMP/NetFlow-style monitoring planned; Packet Tracer validation via CLI show commands.
- Centralized time sync via NTP for consistent log correlation.

This configuration aligns with the hierarchical campus model, addressing plan, device selections, and OSPF-based inter-site design defined in Sections 1 and 2, while meeting the bandwidth and security requirements established in Section 3.

4 Design the network map using Packet Tracer

4.1 The whole system

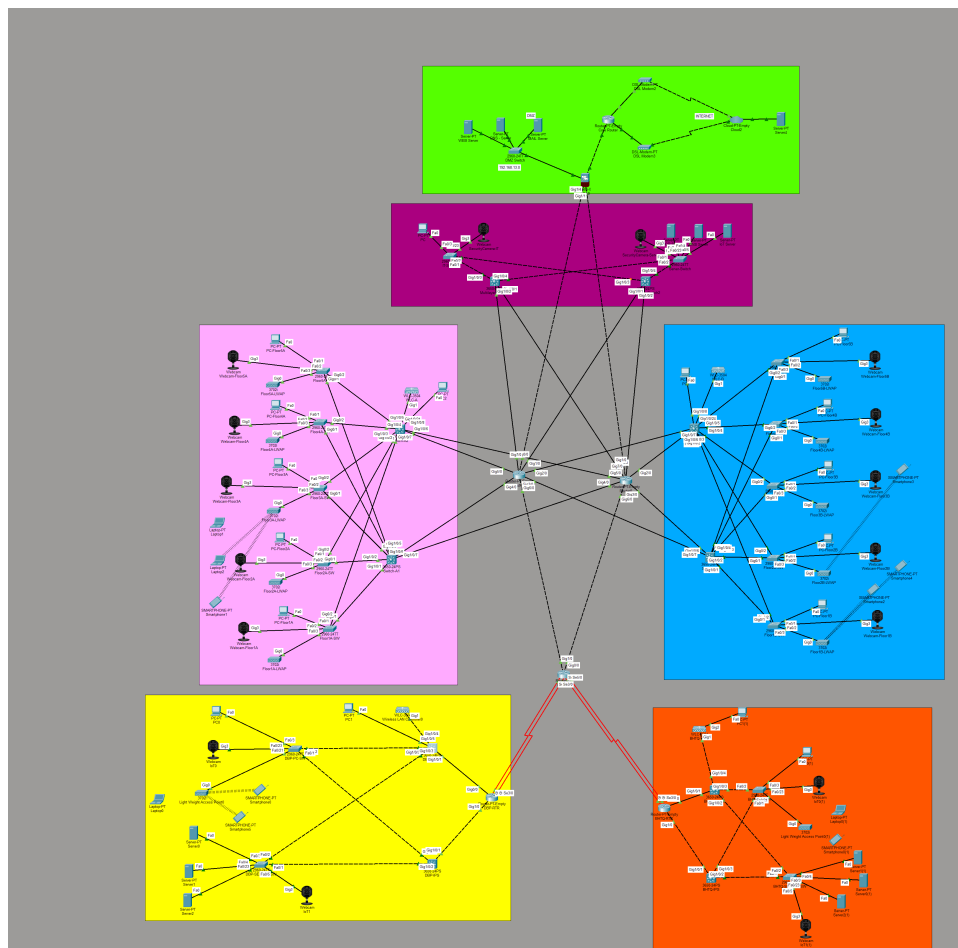


Figure 9: The whole system topology

We have 3 areas of pink, magenta and blue belong to the Main Site. Building A is the pink area, building B is the blue area and the IT block is the magenta one. The two auxiliary sites are DBP (yellow) and BHTQ (orange). Also at the top is our DMZ-Firewall-Internet section (green). We configure our routers and L3 switches using OSPF routing protocol for the whole system. Also We utilized DHCP pool on each switches to assign the IP for end devices. Connections between sites are depend on SD-WAN with leased line using Serial DCE cab.

4.2 DMZ-Firewall-Internet section

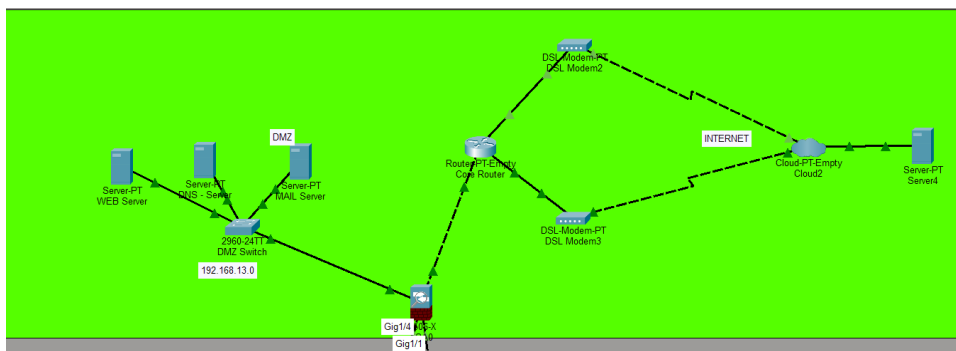


Figure 10: DMZ-Firewall-Internet

4.3 Building A

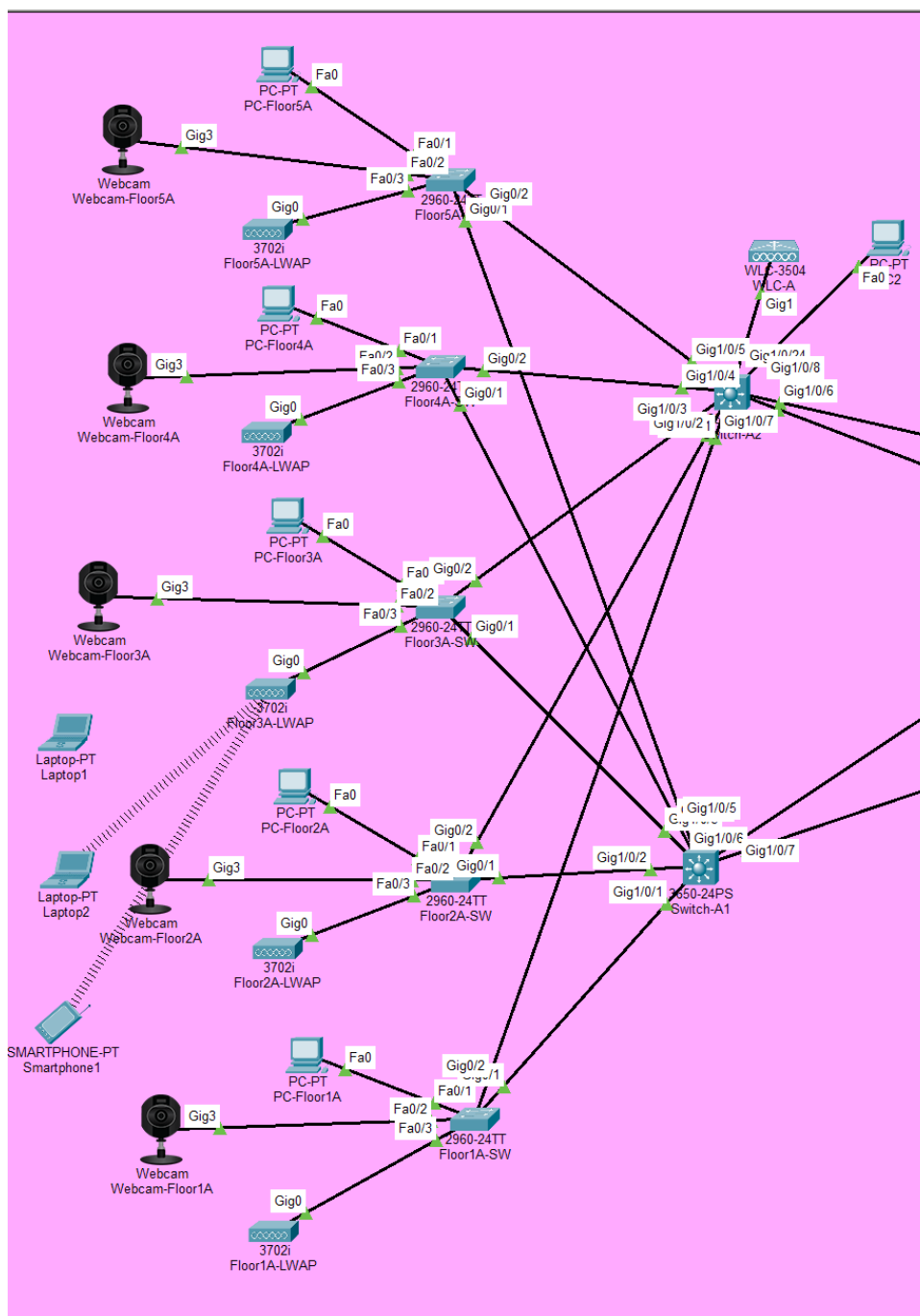


Figure 11: The building A topology

4.4 Building B

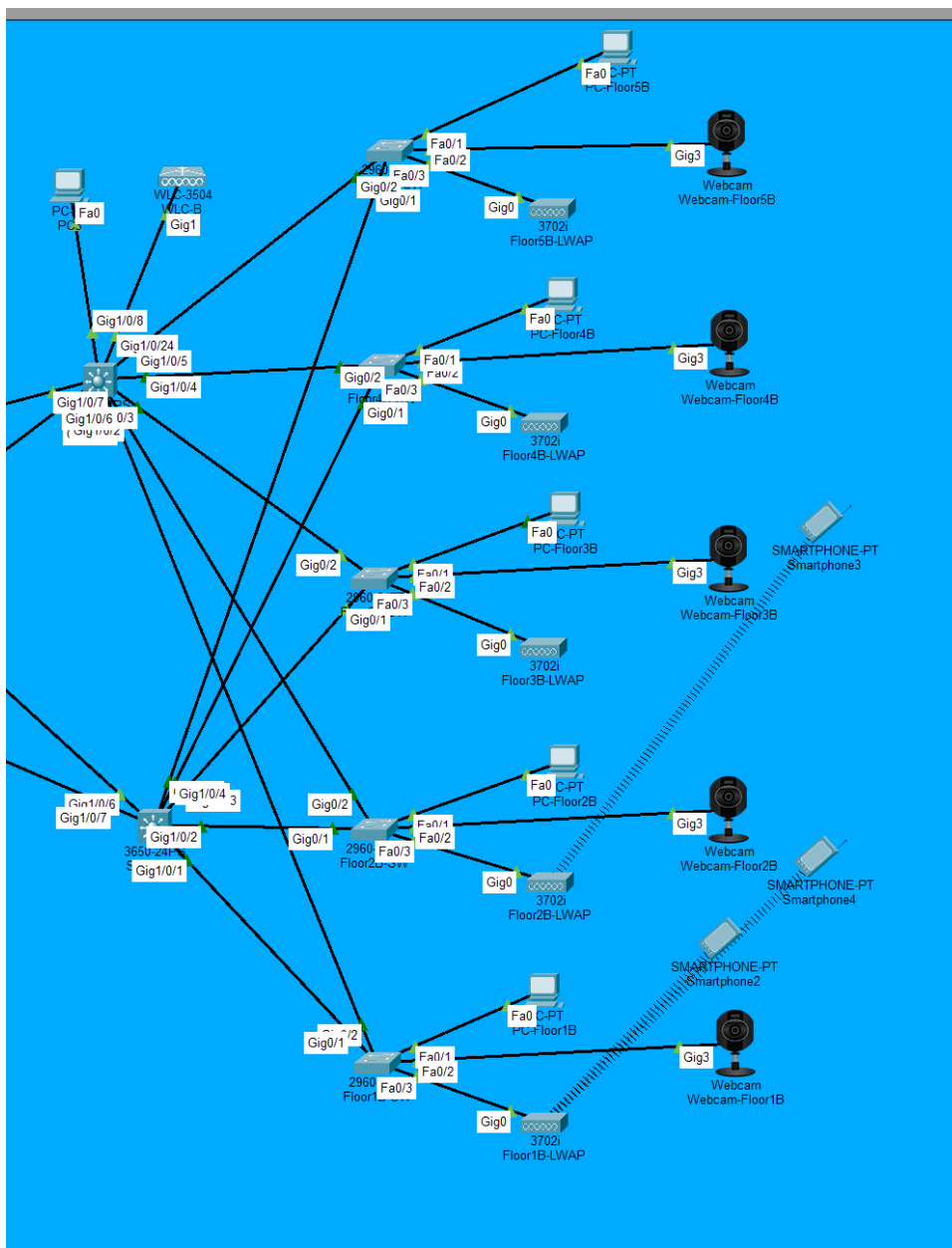


Figure 12: The building B topology

4.5 IT Block



Figure 13: The building BHTQ topology

4.6 Building DBP

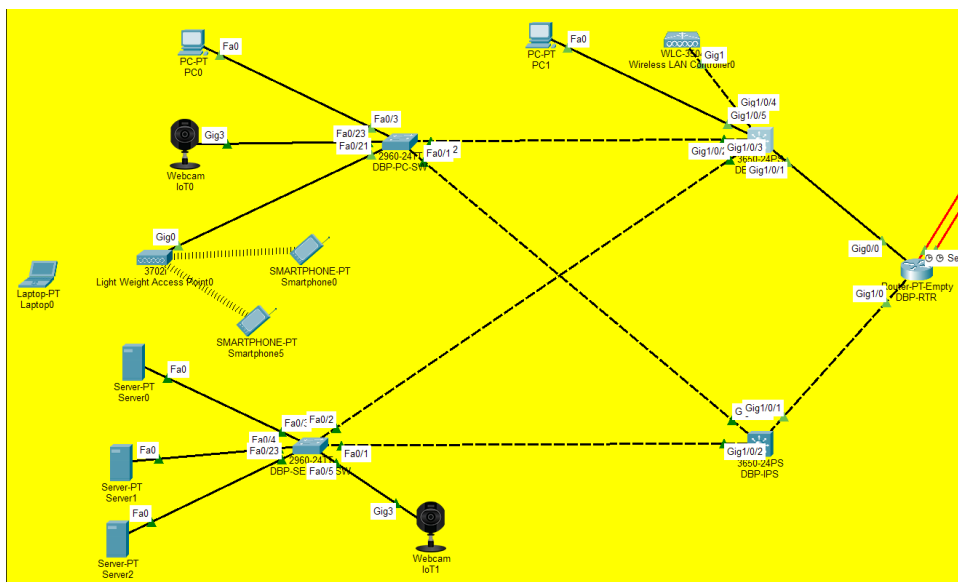


Figure 14: The building DBP topology

4.7 Building BHTQ

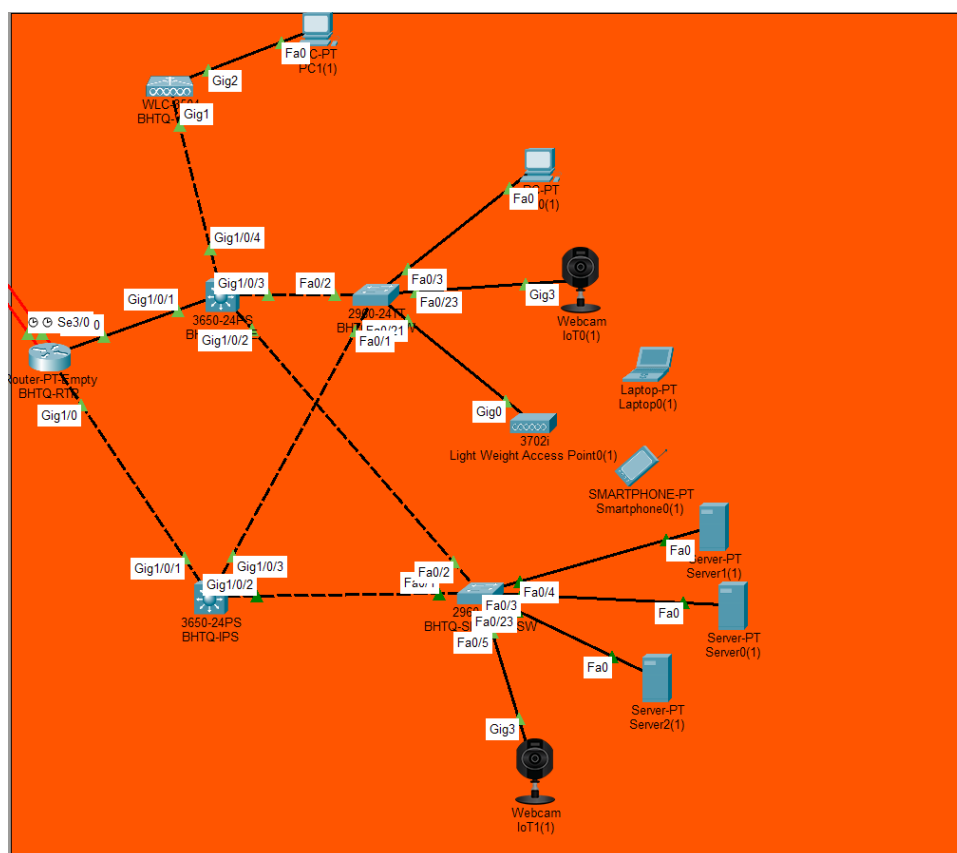


Figure 15: The building BHTQ topology

5 System Verification and Connectivity Tests

5.1 Verifying Dynamic IP Allocation (DHCP)

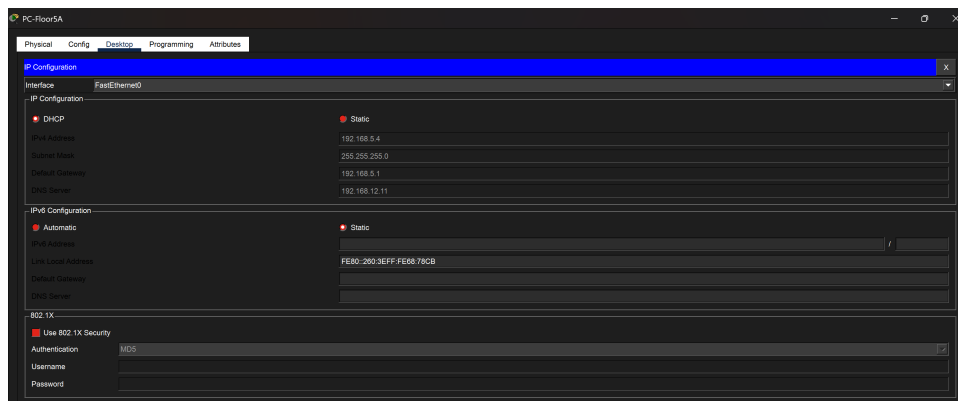


Figure 16: Successful DHCP IP Assignment

Figure 15 demonstrates a PC on the internal network successfully receiving a dynamic IP address from the Layer 3 switch. This confirms that the DHCP pools are correctly configured and operational.

5.2 Testing Intra-VLAN Connectivity (Wireless Peers)

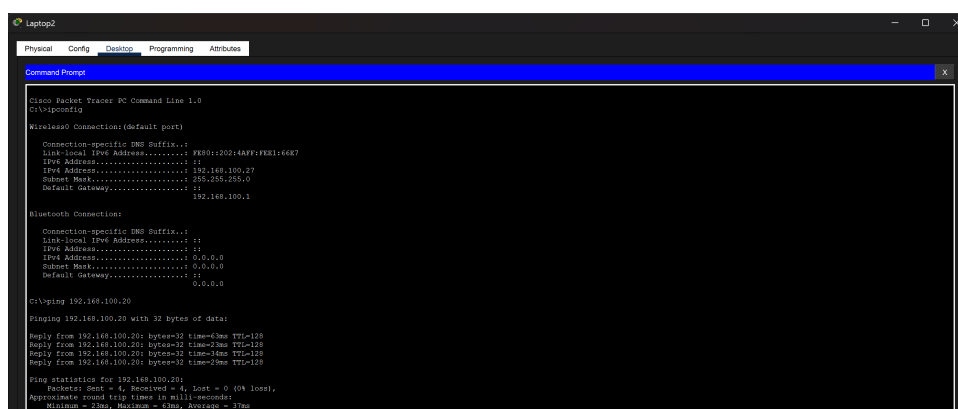


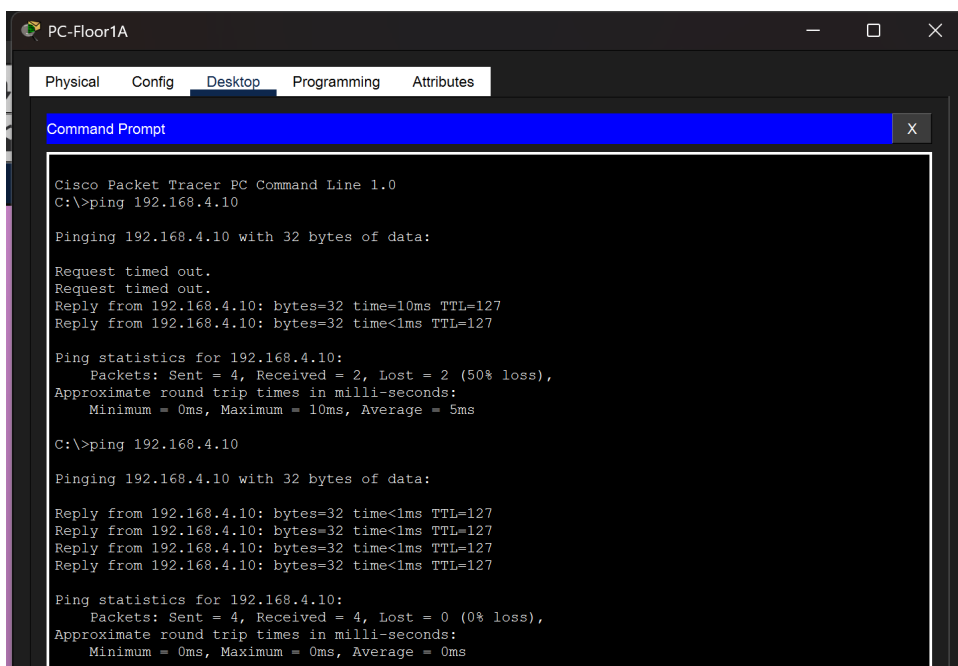
Figure 17: Successful Intra-VLAN Ping (Wireless Clients)

To test connectivity within the same logical network, two wireless clients (Laptop2 and Smartphone1) were connected to the Guest VLAN (VLAN 100). As shown in Figure 10, Laptop2 (IP 192.168.100.27) successfully pings Smartphone1 (IP 192.168.100.20), confirming that client-to-client communication within the wireless VLAN is fully functional.

5.3 Ping devices in different VLANs

5.3.1 Inter-VLAN Routing Test (Within Building A)

This test verifies that the Layer 3 switches (A1, A2) are correctly routing traffic between different VLANs. We ping from PC-Floor1A (VLAN 10) to PC-Floor4A (VLAN 40). As shown in Figure 17, the first ping attempt has partial packet loss while the network discovers the path (ARP), but the subsequent ping is 100 percent successful. This confirms that Inter-VLAN routing is fully operational.



```

PC-Floor1A
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.4.10: bytes=32 time=10ms TTL=127
Reply from 192.168.4.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

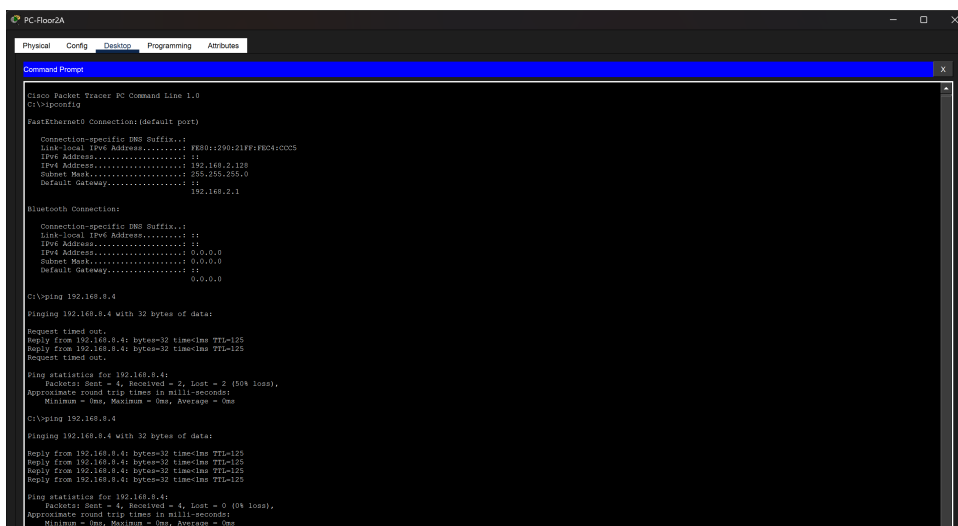
Reply from 192.168.4.10: bytes=32 time<1ms TTL=127
Reply from 192.168.4.10: bytes=32 time<1ms TTL=127
Reply from 192.168.4.10: bytes=32 time<1ms TTL=127
Reply from 192.168.4.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

Figure 18: Ping test from Floor 1A (VLAN 10) to Floor 4A (VLAN 40)

5.3.2 Inter-Building Routing (Main Site)

Figure 18 validates connectivity between the two main campus buildings. A PC in Building A (VLAN 20) successfully pings a PC in Building B (VLAN 30). This confirms that the Core Router (Router0) and the EIGRP/OSPF routing protocol are correctly advertising routes between the separate building distribution blocks.



```

PC-Floor2A
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::290:21FF:FED4:C0C5
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.2.128
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 192.168.8.4

Pinging 192.168.8.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.8.4: bytes=32 time<1ms TTL=125
Reply from 192.168.8.4: bytes=32 time<1ms TTL=125
Request timed out.

Ping statistics for 192.168.8.4:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.8.4

Pinging 192.168.8.4 with 32 bytes of data:

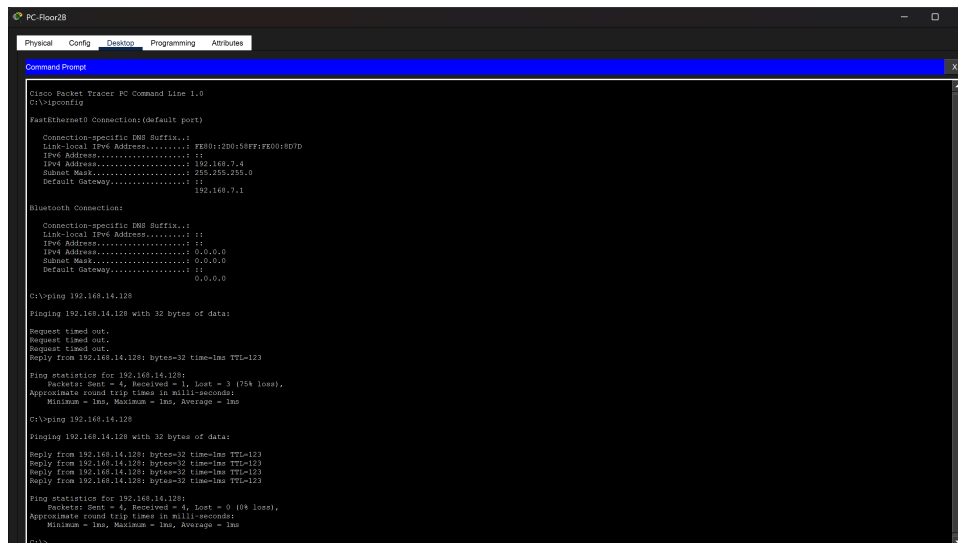
Reply from 192.168.8.4: bytes=32 time<1ms TTL=125
Reply from 192.168.8.4: bytes=32 time<1ms TTL=125
Reply from 192.168.8.4: bytes=32 time<1ms TTL=125
Reply from 192.168.8.4: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.8.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

Figure 19: Inter-Building Ping Test (Building A to Building B)

5.3.3 Inter-Site WAN Routing (Main Site to Auxiliary Site)

This test confirms end-to-end connectivity across the entire WAN. As shown in Figure 19 a PC at the Main Site (Building B) successfully pings a PC at the auxiliary DBP site. This demonstrates that the EIGRP/OSPF routing protocol is fully converged across all routers (Core, A, B, DBP, BHTQ), allowing seamless communication between all locations.



```

PC-Floor2B
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig

C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:56FF:FE00:8D7D
    IPv4 Address. . . . .: 192.168.7.4
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.7.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 192.168.14.128

Pinging 192.168.14.128 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.14.128: bytes=32 time=1ms TTL=123

Ping statistics for 192.168.14.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.14.128

Pinging 192.168.14.128 with 32 bytes of data:
Reply from 192.168.14.128: bytes=32 time=1ms TTL=123
Reply from 192.168.14.128: bytes=32 time=1ms TTL=123
Reply from 192.168.14.128: bytes=32 time=1ms TTL=123
Reply from 192.168.14.128: bytes=32 time=1ms TTL=123

Ping statistics for 192.168.14.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>

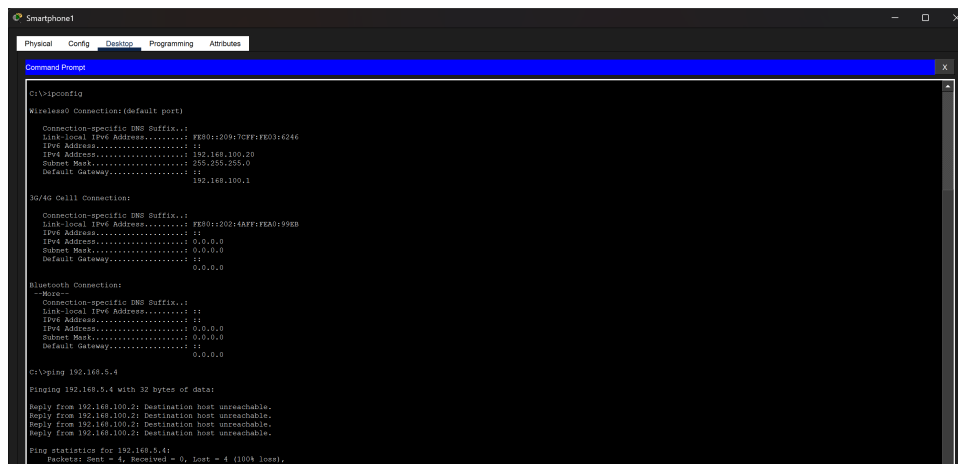
```

Figure 20: WAN Ping Test (Building B to DBP Site)

5.4 Wireless Network Segmentation Test

This section verifies the security policy that isolates wireless devices from the internal wired LAN.

5.4.1 Building A Test



```

Smartphone1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig

C:\>ipconfig

Wireless0 Connection (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::209:7CF7:FE03:6246
    IPv4 Address. . . . .: 192.168.100.29
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.100.1

3G/4G Cell Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::202:6AFF:FEA0:9928
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .:
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 192.168.5.4

Pinging 192.168.5.4 with 32 bytes of data:
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.

Ping statistics for 192.168.5.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Figure 21: Ping failure from Building A wireless network to an internal PC.

A smartphone on the wireless network (VLAN 100, 192.168.100.x) attempted to ping an internal PC (VLAN 1, 192.168.1.4). The test failed as the gateway (192.168.100.1) blocked the connection, returning "Destination host unreachable."

5.4.2 DBP Site Test

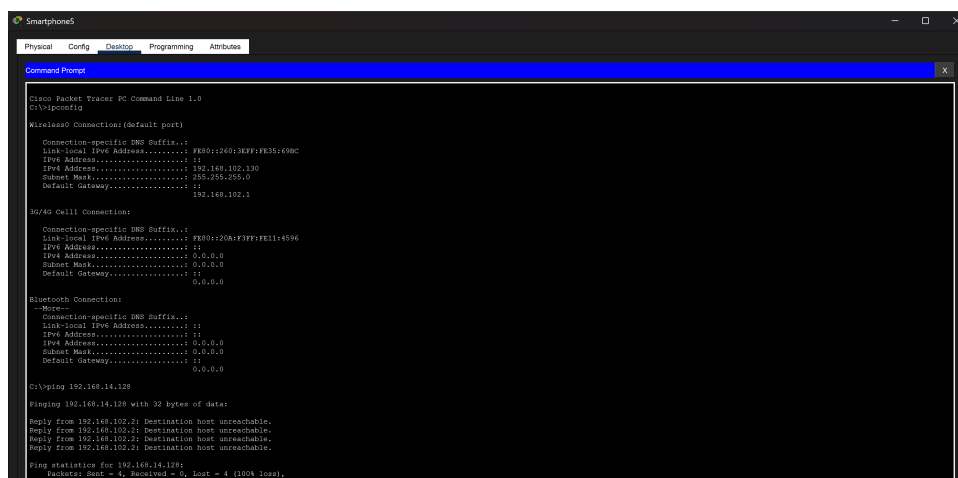


Figure 22: Ping failure from DBP wireless network to an internal PC.

Similarly, a smartphone on the DBP wireless network (VLAN 102, 192.168.102.x) attempted to ping a PC on the DBP internal network (VLAN 14, 192.168.14.x). The gateway (192.168.102.1) successfully blocked this connection.

Both tests confirm that the wireless isolation policy is correctly implemented and enforced.

5.5 Testing Internet Connectivity



Figure 23: Using PC on building A to access Internet.

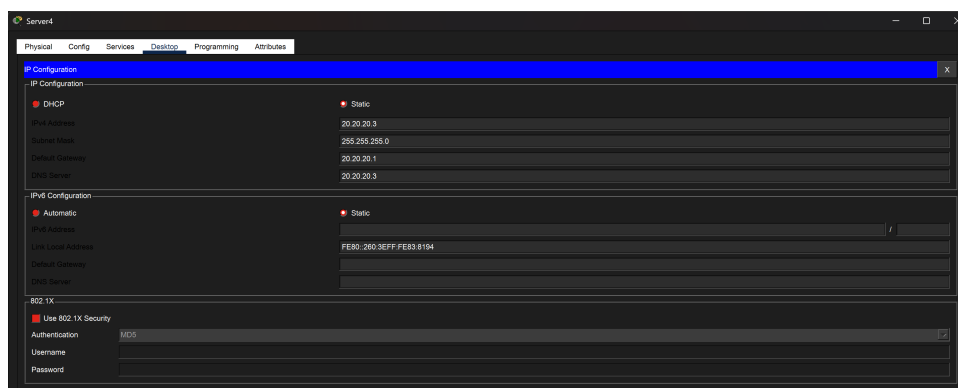


Figure 24: Server IP



In this test, we use a random PC to ping the Internet.



6 Evaluation, Limitations, and Future Development

6.1 Detailed Evaluation of Designed Network Features

Feature	Detailed Evaluation and Status	Justification & Precision
Reliability and High Availability	Medium. The logical three-layer design (Core-Distribution-Access) and the separation into one Main Site and two auxiliary sites provide stable day-to-day operation, but several components are still single points of failure.	The use of dual Layer-3 switches (Multilayer Switch1/2) with HSRP (Hot Standby Router Protocol) provides gateway redundancy for all internal VLANs; if one switch fails, the default gateway at 192.168.x.1 remains reachable. Weakness: The core router and the single ASA firewall remain Single Points of Failure (SPOFs), so a failure of either device may cause a complete loss of WAN or Internet connectivity.
Ease of Upgrade and Scalability	Very potential. The hierarchical design (Core / Distribution / Access) offers a clear and predictable path for future expansion.	The implemented hierarchical design (Core/Distribution/Access) provides a clear path for expansion. New access switches, floors, or even auxiliary sites can be added without modifying the core routing logic. EIGRP automatically learns new networks once their interfaces and VLANs are configured. The consistent use of /24 subnets (253 usable IP addresses per VLAN) leaves ample address space for an estimated 20% growth over the next five years, while keeping addressing and documentation simple.
Network Security and Safety	Basic, perimeter-focused security that is acceptable for a teaching environment but not yet enterprise-grade.	Strengths: Traffic is separated into IT, Server, WLAN, and SecurityCamera segments using VLANs, and the Cisco ASA firewall enforces NAT, stateful inspection, and basic access control between Inside, DMZ, and Outside zones. This prevents direct exposure of internal addresses and limits east-west movement across segments. Weaknesses: The design still relies mainly on static ACLs and VLAN separation; there is no intrusion prevention system, centralized logging, or protection against insider threats and advanced attacks.
Diverse Support and Performance	Good. The network integrates a realistic set of services and technologies that support both operation and management.	The infrastructure combines modern physical media (e.g., GPON backbone, Gigabit Ethernet uplinks) with a structured VLAN design. DHCP and DNS are centralized on dedicated servers so that address assignment and name resolution are consistent across all VLANs and sites. Additional servers emulate web, IoT, and camera services, enabling end-to-end testing of routing, NAT, and security policies and simplifying troubleshooting.

Table 7: Detailed Evaluation of Designed Network Features

6.2 The Remaining Problems and Outstanding Issues

The simulation and design process revealed several practical and functional deficiencies that compromise enterprise reliability:

Problem Area	Specific Impact/Issue	Reference
Core SPOF	Total network failure if the single Core Router or the single ASA firewall fails. The hospital would lose all inter-site WAN communication and all Internet/DMZ access until manual recovery.	Main Site design and checklist where all six Layer-3 switches are connected to one router and one ASA.
Host Security	The system is vulnerable to internal host threats, such as a rogue client performing ARP poisoning or a compromised server providing fake DHCP/DNS information. Current controls mainly protect VLAN borders, not traffic between endpoints in the same VLAN.	Security discussion and test results that only verify ACL-based isolation between Guest WiFi and LAN PCs.
NAT Instability	Dynamic NAT on the ASA is not fully stable in the current simulation. During testing, inside clients could not reach the Internet until NAT was temporarily changed to a static rule and then re-configured as dynamic again. This behavior threatens availability of external services such as web and DNS.	NAT configuration and test scenario in the “Network Address Translation (NAT)” verification part and the NAT subsection in the re-evaluation chapter, where the strange behavior of dynamic NAT is documented.
Security Depth	The security model mainly relies on default ASA behavior (higher-to-lower security levels allowed) and a small number of ACLs. There is no IDS/IPS, rate-limiting, or DDoS protection; large-scale scans or volumetric attacks from the Internet or from compromised internal hosts cannot be realistically detected or mitigated in the current design.	Security subsection of the re-evaluation chapter, and the limitation note that Packet Tracer cannot realistically emulate DDoS or advanced attack scenarios; ASA and ACL configuration sections only show basic filtering.
WAN Load Balancing	While EIGRP is implemented, the required load balancing mechanism across the four leased lines needs verification and fine-tuning to ensure efficient, symmetrical bandwidth utilization.	WAN connection design (leased-line description and EIGRP configuration for Main-DBP and Main-BHTQ links) and inter-site connectivity tests, where reachability is confirmed but throughput and per-link utilization has not yet been measured.

Table 8: Remaining Problems and Outstanding Issues

6.3 Development Orientation in the Future

Future development must focus on layered redundancy, stable application delivery, and automation to achieve a resilient infrastructure suitable for a critical large hospital.

Development Goal	Specific Action Points	Expected Benefit
Achieve Core Redundancy	Deploy Second Core Router (R1): Implement HSRP or VRRP between R0 and R1 to create a non-stop, highly available gateway for the entire network. Add Redundant Firewall: Implement a second ASA in Active/Standby Failover mode to ensure zero downtime at the network perimeter.	Eliminates the single point of failure for Internet and WAN access, significantly increasing overall hospital uptime and ensuring that critical clinical systems remain reachable during hardware failures or maintenance windows.
Implement Zero-Trust Access Security	Enable DHCP Snooping on all access-layer switches (2960) to block rogue DHCP servers and preserve the integrity of IP addressing. Configure Dynamic ARP Inspection (DAI) to mitigate ARP spoofing and man-in-the-middle attacks. Optionally, combine these with Port Security and 802.1X (where supported) to restrict access to authenticated endpoints only.	Protects the network from internal threats by ensuring that only legitimate IP-MAC bindings and authorized devices can communicate, greatly reducing the risk of lateral movement and data tampering from compromised hosts.
Secure Remote Connectivity	Configure IPsec VPN tunnels for site-to-site connectivity between the Main Site and the auxiliary sites (DBP and BHTQ), so that existing leased lines are logically encapsulated inside encrypted tunnels. Optionally extend the design with remote-access VPN profiles for doctors or administrators who need secure off-site access to hospital systems.	Provides secure, encrypted communication for sensitive patient and administrative data over the WAN, aligning the design with confidentiality and compliance requirements while still allowing flexible inter-site and remote access.
Application and Service Maturity	Finalize NAT: Troubleshoot and stabilize the dynamic NAT configuration on the ASA to remove the need for manual resets and ensure consistent outbound connectivity. Define and deploy end-to-end QoS policies that prioritize latency-sensitive applications (VoIP, medical monitoring, critical EMR access) over best-effort traffic such as bulk file transfers and backups.	Ensures that core hospital applications always have predictable connectivity and performance, reducing delays in clinical workflows and improving user experience during peak usage periods.
Centralized Monitoring and Automation	Introduce a centralized monitoring and logging platform (e.g., Syslog/NMS server in the IT block) and configure all routers, switches, firewalls, and servers to send logs and SNMP/telemetry data to it. Standardize configuration templates for recurring components (access switches, branch routers, SSIDs) to simplify deployment and future changes.	Provides better visibility into faults, security events, and capacity trends, enabling faster troubleshooting and proactive maintenance. Template-based configuration and centralized monitoring reduce human error and operational overhead, supporting long-term scalability and manageability of the hospital network.

Table 9: Development Orientation in the Future



7 Resources

The source code is in this Github repository: [link](#)